

Combinatorics PhD Exam, September 2002

SHOW ALL WORK TO RECEIVE CREDIT!!

1. All n soldiers of a military squadron stand in a line. The officer in charge splits the line at several places, forming smaller (nonempty) units. Then he chooses a (possibly empty) subset of the newly formed units for night duty. In how many different ways can he do this?
2. All n soldiers of a military squadron stand in a line. The officer in charge splits the line at several places, forming smaller (nonempty) units. Then he names one person in each unit to be the commander of that unit. Let h_n be the number of ways he can do this, and let $H(x)$ be the ordinary generating function of the numbers h_n . Find $H(x)$.
3. Let G be a simple graph, and let A be the adjacency matrix of G . Decide whether the following statements are true or false.
 - a. A has only real eigenvalues.
 - b. The sum of the eigenvalues of A is 0.
 - c. The determinant of A is always positive.
4. Prove that the number of ways to partition a convex $n + 1$ -gon into triangles and one quadrilateral by noncrossing diagonals is $\binom{2n-3}{n-3}$.
5. Decide whether B_n , Π_n , and NC_n are distributive lattices. (The Boolean algebra, the partition lattice, and the lattice of noncrossing partitions).
6. Let $f_k(n)$ be the number of permutations of length n having k valleys. Is it true that $f_k(n)$ is a p -recursive function of n , no matter what k is?
7. How many permutations of length 8 have descent set $\{1, 4, 6\}$?
8. Let $p = p_1 p_2 \cdots p_n$ be an n -permutation, and assume $n \geq 3$. We say that i is an excedance of p if $p_i > i$. Find the number of n -permutations whose set of excedances is $\{n - 2, n - 1\}$.
9. Let D be the partially ordered set of positive integers ordered by divisibility. Find a formula for $\mu(1, x)$.
10. Find an explicit formula for the numbers a_n , if $a_{n+1} = (n + 1)a_n + 2(n + 1)!$ if $n \geq 1$, and $a_0 = 0$.